

Finite Difference Modeling of Heat Transfer in Building Wall Insulation

Numerical Analysis (MATH 306)

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Introduction

Heat transfer plays a critical role in the energy analysis of any object. Thus, understanding its characteristics and dynamics open opportunities for energy optimization in the daily life. One important heat transfer application is the wall insulation which is used to control the temperature gradient between inner and outer surfaces. While the Analytical approach is used for simple geometries and conditions, it falls short when the temperature changes with more than one variable. Since most of the real-world scenarios are driven by many variables (e.g., material properties, geometry, boundary conditions), the numerical modelling was needed to account for them.

The aim of this report is to apply the **Finite Difference Method (FDM)** to simulate the conduction heat transfer at steady state through a 2D cross-section of an insulated wall. This report will show how FDM can be used to estimate the temperature distribution across the wall. To fully define the simulation, fixed temperatures on the boundaries will be used. The simulation will be made using Python to utilize its built-in visualization functions.

Problem Description

We consider a section of a wall with a unity thickness and homogeneous material properties. The temperature are assumed to be fixed and at a higher temperature on the interior side of the wall, while the exterior side is fixed at a lower temperature. Additionally, the top and bottom boundaries are assumed to have a fixed temperature difference. This leads to the following physical assumptions:

- **2D steady-state heat conduction** (no time dependence)
- **Constant thermal conductivity**
- **Dirichlet boundary conditions (**fixed variable value**)** on all four edges

The mathematical formulation of this physical model is based on the **Laplace equation**:

$$\frac{\partial^2 T}{\partial x^2} + \frac{\partial^2 T}{\partial y^2} = 0$$

Where $T(x, y)$ is the temperature distribution in the wall cross-section.

Mathematical Formulation

The governing equation for steady-state heat conduction in two dimensions (with no internal heat generation) is:

$$\frac{\partial^2 T}{\partial x^2} + \frac{\partial^2 T}{\partial y^2} = 0$$

The domain is rectangular, defined over $0 \leq x \leq L_x$ and $0 \leq y \leq L_y$, and discretized into a grid of points.

Boundary Conditions

- **Left boundary ($x = 0$):** $T = T_{x=0} = 100℃$
- **Right boundary ($x = L_x$):** $T = T_{x=L_x} = 0℃$
- **Bottom boundary ($y = 0$):** $T = T_{y=0} = 25℃$
- **Top boundary ($y = L_y$):** $T = T_{y=L_y} = 75℃$

These conditions lead to a **2D** temperature distribution across the wall.

Numerical Methodology

To accurately demonstrate the temperature behaviour using Laplace equation, the domain is discretized into a 2D Cartesian grid with uniform spacing h in both x and y directions. The Finite Difference Method (FDM) is applied using a 5-point stencil.

Discretization

The discrete approximation at interior nodes is:

$$T_{i,j} = \frac{(T_{i+1,j} + T_{i-1,j} + T_{i,j+1} + T_{i,j-1})}{4}$$

Boundary Treatment

All four boundaries use **Dirichlet conditions**.

Iterative Solver: Successive Over-Relaxation (SOR)

The SOR method accelerates convergence using Gauss-Seidel. The update rule is:

$$T_{i,j}^{(k+1)} = \frac{(T_{i+1,j}^{(k)} + T_{i-1,j}^{(k+1)} + T_{i,j+1}^{(k)} + T_{i,j-1}^{(k+1)})}{4}\lambda + (1-\lambda)T_{i,j}^{(k)}$$

with a relaxation factor $1 < \lambda < 2$.

Convergence Criterion

Iteration continues until:

$$\max_{i,j}\left|T_{i,j}^{(k+1)} - T_{i,j}^{(k)}\right| < \epsilon \qquad \text{with } \epsilon = 10^{-6}$$

Python Implementation

```
import numpy as np
import matplotlib.pyplot as plt

# Domain parameters
Lx, Ly = 1.0, 1.0  # Wall dimensions in meters
Nx, Ny = 100, 100  # Number of grid points
dx = Lx / (Nx - 1)
dy = Ly / (Ny - 1)
Lambda = 1.8  # Relaxation factor
tolerance = 1e-6
max_iter = 10000

# Temperature boundary conditions
T_x_0 = 100.0  # x=0
T_x_Lx = 0.0  # x=L_x
T_y_0 = 25.0   # y=0
T_y_Ly = 75.0  # y=L_y

# Initialize temperature grid
T = np.zeros((Ny, Nx))
T[:, 0] = T_x_0
T[:, -1] = T_x_Lx
T[0, :] = T_y_0
T[-1, :] = T_y_Ly

# Iterative solver: SOR
for iteration in range(max_iter):
    T_old = T.copy()

    for j in range(1, Ny - 1):
        for i in range(1, Nx - 1):
            T[j, i] = (1 - Lambda) * T[j, i] + (Lambda / 4) * (
                    T[j, i + 1] + T[j, i - 1] + T[j + 1, i] + T[j - 1, i]
            )

    # Neumann BCs: zero flux (top and bottom)
    T[0, :] = T_x_0
    T[-1, :] = T_x_Lx
    T[0, :] = T_y_0
```

```
    T[-1, :] = T_y_Ly

    # Convergence check
    if np.max(np.abs(T - T_old)) < tolerance:
        print(f'Converged in {iteration} iterations.')
        break

# Plot results
plt.figure(figsize=(8, 6))
cp = plt.contourf(T, 100, cmap='inferno')
plt.title('Temperature Distribution in Insulated Wall')
plt.xlabel('Width (x)')
plt.ylabel('Height (y)')
plt.colorbar(cp, label='Temperature (°C)')
plt.show()
```

Results and Discussion

The simulation produced a two-dimensional temperature distribution due to the variation in both x and y directions. The temperature field shows diagonal gradients and curved isothermal (constant temperature) lines, as heat flowed from the hot left boundary and top wall toward the cold right boundary and cooler bottom surface.

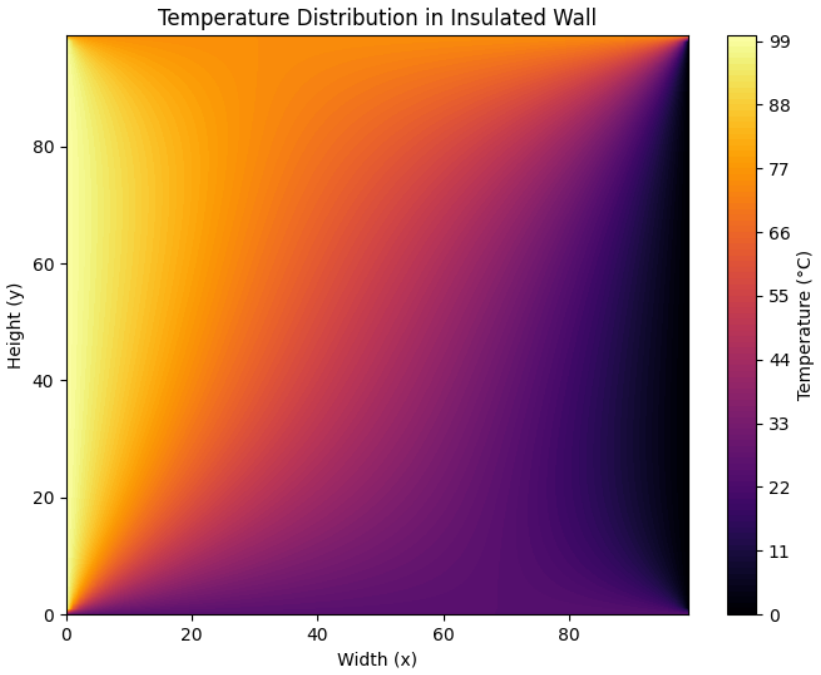


Figure 1: Simulated temperature distribution across the wall using the Finite Difference Method.

Observations:

- Vertical and horizontal gradients are present.
- Using more grid points (lower h value) yields to smoother, more accurate profiles.
- A relaxation factor $\lambda = 1.8$ significantly speeds up convergence.

Conclusion

This project demonstrated the application of the FDM to solve a 2D steady-state heat conduction problem with fully defined boundary temperatures. By using Dirichlet conditions on all edges and a discretized grid, a temperature field was obtained.

The approach is general and can be easily extended to more complex real-life applications, providing a strong aid for optimization goals.

References

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3. Incropera, F. P., & DeWitt, D. P. (2006). *Fundamentals of Heat and Mass Transfer*. Wiley.